How far does a learned surrogate of a flowsheet simulator transfer across ores? A closed-circuit, size-by-mineral engine, with cases and ore groups held out

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Technical report · v0.10.008, 1 November 2026 · Licence CC BY 4.0
CAOS open-research programme · mineral-processing simulator
Version DOI 10.5281/zenodo.23850020 · concept DOI (always latest) 10.5281/zenodo.23783845
Not peer reviewed. Source code, data and computational records:
github.com/fsantibanezleal/CAOS_OreFlow

Abstract

Learned surrogates of process simulators are often evaluated on random states from simulated plants they have already seen. We compare interpolation, leave-one-case-out and leave-one-ore-group-out evaluation for five models trained on a steady-state mineral-processing simulator. The simulator tracks mineral mass over 63 size classes, water and closed grinding circuits, with unit-level balance audits and explicit misses against published examples. On a 3,072-state design over twelve authored cases, the multilayer perceptron has an interpolation recovery RMSE of 3.38 percentage points. Its magnetite transfer forecast is wholly outside a declared gross-range diagnostic and remains a rejected fold with null error metrics. Over the eleven case folds admissible for all models, mean recovery RMSE is 6.86 points for the perceptron, 8.67 for gradient boosting and 9.40 for the Gaussian process. These conditional results do not describe performance on the excluded case. Four of five neural seeds reject magnetite; the remaining seed retains a 217.18-point error, exposing instability that an interpolation score conceals. The perceptron's recovery R^2 is 0.795 on five chalcopyrite cases held out together and -10.2 on two gold-in-pyrite cases. Group-held-out log-upgrade median R^2 is negative for every model. Results therefore depend on the target, held-out unit, admissibility diagnostic and cohort denominator. The cases are authored; these results do not establish performance on operating plants.

1 Introduction

A surrogate of a flowsheet simulator is attractive for the same reason as any surrogate: the simulator is slow and the surrogate is fast, so it can sit inside an optimizer, a sweep or an interactive tool. Whether it can stand in for the simulator depends on the question it is asked. Inside the envelope it was trained on, an interpolation score answers that question. A digital twin is asked something harder: what happens with a different ore, a harder or finer feed, another plant. That is transfer, and an interpolation score says nothing about it.

This work measures both on the same models, with a simulator whose answers are known exactly and can be reproduced. Its contributions are three:

1. an open flowsheet engine that represents streams by size and mineral, solves closed circuits, audits every balance from its output streams, is checked against published examples with every miss stated, and is reproduced by a line-by-line second implementation;
2. a test protocol that scores learned surrogates of it by interpolation, by leaving one case out and by leaving one ore group out over twelve authored plants in four circuit families, with interval coverage and an out-of-envelope guard;

3. a measured transfer failure and seed instability on magnetite, conditional comparisons that retain the failed fold, distinct results for whole-case and ore-group holdouts, and a reconstruction guard whose missed feature shifts remain visible by input and distance.

2 Process model

The record reproduces all 96 committed states within 1e-6 in the browser port and all 96 committed states close within 1e-9 at every unit in the Python engine. The port shares constants and is a translation check, not an independent validation of physical assumptions.

2.1 Representation

Every stream holds, for each mineral, a vector of mass flows (t/h) over one grid of 63 size classes on a fourth-root-of-two progression from 150 mm to a pan below 3.24 µm, and a water flow. Assays are computed from mineral masses and the minerals' element contents (from their formulas and the IUPAC atomic weights); no grade is stored, so a grade changes only when a mass does. Liberation is described per valuable mineral by size (King 1979): a liberated fraction, and composites whose content and host are declared.

2.2 Comminution

The crusher is the Whiten form (Duarte et al. 2021), with a strictly lower-triangular breakage matrix $\mathbf{B}$ and a classification function $\mathbf{C}$ set by the closed-side setting:

$$\mathbf{p} = (\mathbf{I} - \mathbf{C})(\mathbf{I} - \mathbf{B}\mathbf{C})^{-1}\mathbf{f}$$

The ball mill is an energy-specific population balance (Herbst and Fuerstenau 1980) in the form of the Moly-Cop tools: three perfect mixers of volume fractions 0.70, 0.15 and 0.15 share the breakage operator $\mathbf{D} = (\mathbf{I} - \mathbf{B})\,\mathrm{diag}(S^E)$, so the mill's inverse transfer at an energy per pass e is a cubic polynomial in $\mathbf{D}$, and the closed circuit with the cyclone returning a fraction C_i of each class solves

$$\left[\mathbf{T}^{-1}(e) - \mathrm{diag}(C)\right]\mathbf{p} = \mathbf{f},$$
$$\mathbf{T}^{-1}(e) = \mathbf{I} + c_1 e\mathbf{D} + c_2 e^2\mathbf{D}^2 + c_3 e^3\mathbf{D}^3$$

per mineral. Two scalar unknowns, the energy per pass that meets the target P80 and the host cut that holds the design circulating load, are found by the Illinois method. The energy per pass is per tonne of mill feed, so the mill draws $e\,m$, with m the fresh feed plus the cyclone underflow (less what a gravity bleed takes from it); when that exceeds the installed power, the circuit runs at installed power, $e\,m(e) = P_{\mathrm{inst}}$ solved as a fixed point, with a coarser product, and says so. The reported grinding energy is that net energy per tonne of fresh feed, $\frac{e\,m}{F}$; the crushing energy is Bond's with the crushing work index, and the Bond requirement $E = 10W_i\left(\frac{1}{\sqrt{P}} - \frac{1}{\sqrt{F}}\right)$ gives the operating work index and the efficiency ratio beside it, with Rittinger and Kick calibrated to agree with Bond at a reference reduction and never added to it (Bond 1952; GMG 2021).

2.3 Classification and separation

Hydrocyclones follow Plitt (1976): a Rosin-Rammler partition

$$y(d) = R_f + (1 - R_f)\left[1 - \exp\left(-\ln 2\left(\frac{d}{d_{50c}}\right)^m\right)\right]$$

with a water bypass R_f from the water balance and a cut corrected for each mineral's density, so dense minerals classify as coarser particles and circulate. Plitt's sizing of the cyclone cluster is uncalibrated and fails its published check (section 3), so the cluster's count and pressure are reported as an estimate, never as a result. Flotation banks are perfect mixers in series; the rate constant of each particle class follows from the bubble surface area flux $S_b = \frac{6J_g}{D_{32}}$ (Gorain et al. 1997, 1999) and a size window (Trahar 1981), a composite floats at the liberated rate scaled by its exposed surface, $P_{\text{comp}} = P_V \phi^{\frac{2}{3}}$ with ϕ the valuable mineral's volume share, entrainment follows Savassi et al. (1998), and cleaner and recleaner tails return to the preceding bank. Each cell's residence is its pulp volume, less the gas hold-up, over the flow that leaves it as tail, which depends on what the cell floats, so it is solved per cell as a fixed point:

$$
\begin{aligned}
r_j &= \frac{k\tau_j + \mathrm{ENT}\, w_j}{1 + k\tau_j + \mathrm{ENT}\, w_j}, \\
\tau_j &= \frac{V(1-\varepsilon_g)}{Q_{\text{tail},j}}, \\
R_N &= 1 - \prod_j (1 - r_j)
\end{aligned}
$$

Gold follows the gravity-recoverable gold (GRG) model of Laplante, Woodcock and Noaparast as Vincent (1997) states it: the gold is a GRG component with its own size distribution, from a GRG test, plus a non-GRG component; the GRG's cyclone cut takes a density correction with an exponent of 1.0, fitted to measured GRG partitions where Stokes would give 0.5; the gravity unit on a bleed of the cyclone underflow, or of the mill discharge, recovers GRG by size at a per-pass recovery $R(d) = R_{\max}\left[1 - \exp\left(-\left(\frac{d}{x_g}\right)^2\right)\right]$; and grinding moves GRG to non-GRG at a slowed breakage rate (Banisi et al.). Low-intensity magnetic drums capture liberated magnetite and composites by their magnetite content; desliming cyclones discard slimes ahead of flotation, with the water split following the solids sent down at a declared underflow density.

2.4 Balance closure and the contract

An independent recomputation calculates, from the output streams alone, the closure of every mineral, every species and the water at every unit and for the circuit, and of every size class at the units that do not break particles. The operating point is thirteen inputs declared once, with units, per-case bounds, steps, the families they apply to, a cross-field rule and bilingual messages; the exported declaration is interpreted identically by the engine's validator, the web service and the browser, so a state is accepted or rejected with the same code everywhere. A state the engine cannot solve, such as a classifier cut at which the mill cannot draw its installed power, is refused with a reason rather than answered.

3 Verification

Balances. All 96 committed states close within 1e-9 relative at every unit and for the circuit, as recomputed from the stored streams by the artifact checks.

The second implementation. A line-by-line TypeScript port runs the engine in the browser. Re-simulating every committed state from its own definition, it matches every metric, stream record, curve and kinetic record within 1e-6 relative, with the same flags. It is a translation by the same author that reads the same constant, atomic-weight and mineral files, so it catches translation errors and not modelling errors. An independent verification of the 0.08.001 implementation against its equations, one reviewer per part of the engine and every finding verified, found defects in both engines, among them the composite rate, the cell residence, the mill power basis and the desliming water split; each was fixed, the engine's in both languages, with a test that fails on the earlier code, and every number below is from the fixed engine.

Published examples. Each is labelled as a published example, not as plant data:

Example	Published	Engine
Moly-Cop BallSim base case (504 t/h, every published input)	net 7.71 kWh/t, gross 8.56; cut 183.3 µm; 6 cyclones at 53 kPa	net 7.30 kWh/t, 5.2% below, inside the 20% tolerance set before the first run; cut 188 µm; Plitt's uncalibrated sizing asks for 2 cyclones at 816 kPa
GMG worked examples of the operating work index	14.4 and 11.7 kWh/t	14.38 and 11.71 kWh/t
Laplante and Staunton gravity example, run like for like (80.4% GRG, the unit on 10 to 60% of the mill discharge)	GRG recovery 79.8 to 95.9%, GRG circulating load 2016 to 413%	GRG recovery 69.5 to 90.7%, GRG circulating load 696 to 84%: outside the 5-point and 35% tolerances
Zandrivierspoort magnetite (a different ore; the 45 µm product is a regrind)	64.9 and 69.0% Fe at 75 and 45 µm	63.8 and 67.8% Fe: a step of 4.0 points against 4.1, at levels 1.1 and 1.2 points below

The gravity example misses because the measured GRG of the case's ore (Snip) is very fine: 9 to 31% of the GRG, almost all finer than about 37 µm, leaves by the overflow; without the GRG finer than 25 µm the GRG recovery comes within 1.1 points from the 20% row on, and the circulating load stays 50 to 69% low. The magnetite example checks a direction and a proportion, not a level, because its ore is not the engine's case.

4 Evaluation design

4.1 Authored circuits

Twelve authored cases in four circuit families: nine copper, nickel, zinc and refractory-gold rougher-cleaner flotation circuits (`rougher`), free-milling gold with a gravity bleed (`gravity_rougher`), fine magnetite with magnetic drums instead of flotation (`magnetic`), and phosphate with desliming ahead of flotation (`deslime_rougher`). Every parameter carries its unit and either a source or the label authored; none is fitted to a plant. Five of the twelve are copper sulphide plants on chalcopyrite, and the two gold plants carry most of their gold in pyrite (55%, beside electrum in the free-milling ore and arsenopyrite in the refractory one); the other five each have a payable and dominant carrier no other case shares.

4.2 Design, features and targets

For each case a scrambled Sobol sequence draws 256 states the contract accepts over the case's whole envelope and two ore factors (liberation size and floatability within their uncertainty half-widths), except that magnetite varies liberation alone. There are 3072 states in all; the engine simulates every state. Each state is described by 22 physical features computed before simulation: the payable fraction, throughput per installed megawatt, the grind, circulating load, water, crusher setting and work index, the payable carriers' share-weighted liberation size, composite content, density, floatability and dose ratio, the rougher gas velocity, cells and volume per t/h, the gravity bleed, the desliming cut and four circuit descriptors. The case identity is never a feature. The targets are recovery (%), the log10 upgrade ratio (concentrate over head grade) and the total specific energy (kWh/t).

4.3 Models and protocols

Ridge (alpha 1); a random forest (300 trees; Breiman 2001); histogram gradient boosting (300 iterations; Friedman 2001); a Gaussian process with a squared-exponential kernel with one length scale per feature and a noise term, on a seeded subsample of 500 training rows (Rasmussen and Williams 2006); and a perceptron

of two hidden layers of 64 units, trained by full-batch AdamW with early stopping on a 15% validation split (Loshchilov and Hutter 2019). Features are standardized on the training rows of each split. Perceptron targets are standardized on those same rows; the Gaussian process normalizes its own training responses, while ridge and the tree models fit targets in their declared units. Because the Gaussian process sees fewer rows, ridge, random forest and gradient boosting are also refitted on the same 500 rows for a separate comparison. The perceptron's ordinary scores retain the full training subset. Because one training seed is one draw, the perceptron is retrained with four more seeds for interpolation and each of the twelve case holdouts. The two multi-case ore-group fits use the base seed; no five-seed group-transfer stability claim follows from this study.

The standardization fits the training subset alone. Numerically constant training columns retain unit scale, including columns whose small positive variance comes from summation roundoff. An unseen value is therefore not amplified by division through a spurious near-zero scale. Neither held-out features nor held-out targets determine the fitted transformation. The interpolation comparison also retains a case-mean baseline fitted on each case's training rows, and an R^2 denominator based on within-case variation, separating case recognition from response prediction.

Interpolation holds out 20% of the states inside every case (2460 training, 612 test rows). *Leave one case out* holds out each case in turn; the models that predict a case never saw any of its states, but they may have seen plants on the same ore. *Leave one ore group out* holds out every plant that shares the held-out plant's payable and dominant carrier mineral: the five chalcopyrite plants together (1792 training, 1280 test rows), the two gold plants together (2560 training, 512 test rows), and each of the other five alone, where it is the same fold as leaving one case out; seven groups in all. An autoencoder (16, 6 and 16 units) trained on the features gives a guard: its threshold is the 99th percentile of the reconstruction errors of its validation rows. Its false-alarm rate is measured on the 612 held-out in-envelope states. Its headline false-accept rate uses 11,016 probes, each a held-out state pushed half a training span above the maximum of one continuous feature. Separate acceptance curves repeat those probes at six distances from 0.02 to 1.0 spans, independently above the maximum and below the minimum.

4.4 Prediction diagnostics and admissible comparisons

Shape and finiteness are checked before an error is computed. A further gross-range diagnostic compares the prediction range with the target range over the complete authored design, expanded by one observed span on each side. For reference limits a and b, the diagnostic interval is

$$[a - (b - a),\ b + (b - a)].$$

This is a post-inference diagnostic and includes held-out target values. It is not a training-support test, an operational constraint, an independent transfer validation or a rule for selecting fitted parameters. No prediction is clipped or refitted using these limits. Because the test concerns disjoint ranges, it can miss extreme individual predictions when other predictions overlap the interval; passing it does not establish physical validity at every state.

A finite transfer forecast wholly outside that interval is retained as a rejected transfer, with null RMSE, MAE and R^2, unmodified observed/predicted/reference ranges, affected-row count and model, target, protocol, fold and seed identifiers. The same check applies to every constituent case of a grouped fold, so pooled overlap cannot conceal an entirely invalid case. Nonfinite, shape, fitting and programming failures abort evaluation; interpolation range failures also abort. Finite poor forecasts inside the diagnostic envelope retain their errors and negative R^2. Unexpected constant forecasts are identified separately from the intentionally constant baseline.

Conditional summaries report attempted, scored and rejected counts beside their error statistics. A missing admissible cohort has null metrics. Comparisons across models additionally use the intersection of folds admissible for every model, with all excluded identifiers retained. These matched summaries remain conditional descriptions of that intersection; they do not erase a model's failed transfers or rank performance over the complete planned design. Complete per-fold, per-case and per-seed evidence is retained in the release record.

5 Results

5.1 Interpolation and case holdout

Table 1. Recovery interpolation and conditional leave-one-case-out scores. RMSE is in percentage points. Case counts are scored folds out of twelve attempts; the remaining fold is retained as rejected. Case means, maxima and medians use only each model's scored folds, so their denominators differ.

Model	Interp. RMSE	Interp. R^2	Cases scored/12	Case mean RMSE	Case maximum RMSE	Case median R^2
Ridge	9.46	0.617	12	13.68	23.05	-0.127
Random forest	6.49	0.820	12	13.16	48.86	0.359
Histogram gradient boosting	4.60	0.909	12	12.66	56.61	0.512
Gaussian process	5.34	0.878	12	10.95	27.91	0.481
Perceptron	3.38	0.951	11	6.86	12.76	0.573

The perceptron has the lowest interpolation recovery RMSE (3.38 points) on the common 612-row test set. Its conditional case mean is 6.86 points over eleven scored folds, with one rejected recovery fold. Comparing that value directly with another model's twelve-fold mean would confound model error with cohort selection. Table 3 instead compares the same eleven folds for every model.

The five-seed record gives interpolation RMSE from 3.00 to 3.54 points. The first four seeds each score eleven case folds and reject magnetite; their conditional mean RMSE ranges from 6.86 to 7.63 points. The fifth scores all twelve folds, with a 217.18-point magnetite RMSE and a 25.05-point case mean. The range diagnostic admits that last forecast because its prediction range overlaps the widened reference interval. This very poor finite error remains scored. The five conditional means therefore cannot be interpreted as a stability interval over an identical cohort.

5.2 Case transfer errors

Table 2. Recovery RMSE (points) on each held-out case.

Held-out case	Ridge	Random forest	Gradient boosting	Gaussian process	Perceptron
Soft copper porphyry	12.54	11.59	9.47	11.09	5.98
Hard copper porphyry	9.62	6.72	4.82	5.08	1.72
Free-milling gold with gravity	8.55	12.48	10.53	15.83	12.76
Fine magnetite concentration	21.01	48.86	56.61	27.91	Rejected
Nickel sulphide with serpentine slimes	13.47	8.49	6.80	6.66	6.61
Phosphate with clay slimes	15.85	10.04	10.05	10.83	8.93
Copper-molybdenum bulk flotation	9.38	6.12	4.03	6.33	1.91
Oxide copper: assumed sulphidized response	23.05	14.12	15.19	12.94	11.60
Zinc sulphide	10.87	11.88	10.18	8.49	7.26
Copper ore with clay	10.69	5.99	6.06	9.47	6.57
Low-grade copper at high throughput	9.68	6.25	4.48	6.41	2.23
Refractory gold in sulphides	19.44	15.41	13.77	10.33	9.86

The copper cases are not uniformly easy: holding out the soft porphyry costs the perceptron 6.0 points, while its error on the hard porphyry, molybdenum and low-grade copper cases is 1.7 to 2.2 points. Mixed high-clay

copper is 6.6 points. For all 256 magnetite rows, the base-seed recovery forecast ranges from -195.41 to -127.94%, wholly below the diagnostic interval of -92.94 to 196.38%; the observed simulator recovery is 68.93 to 93.32%. No clipped prediction or replacement error enters Table 2. The tree models also have large finite errors on magnetite. These model errors are measured against the authored simulator, not a plant assay.

Table 3. Matched admissible recovery cohorts. Each case column uses eleven of twelve planned folds; each group column uses six of seven. Magnetite is excluded from both intersections because the neural forecast was rejected. RMSE means are unweighted fold means, in percentage points. These are conditional descriptions, not rankings over the complete planned design.

Model	Case mean RMSE	Case median R^2	Group mean RMSE	Group median R^2
Ridge	13.01	0.013	15.15	-0.638
Random forest	9.92	0.430	11.53	0.209
Histogram gradient boosting	8.67	0.581	10.87	0.365
Gaussian process	9.40	0.498	10.95	0.318
Perceptron	6.86	0.573	14.60	0.455

5.3 Whole-ore-group transfer

Seven effective ore-group folds are planned: two multi-case groups and five single-case groups. The single-case groups use their already fitted case folds. The perceptron's magnetite rejection therefore also appears in group transfer; its group summary scores six folds, while the other models score seven.

Table 4. Conditional ore-group recovery scores. RMSE is in percentage points; counts are scored groups out of seven attempts. The perceptron's one rejected group remains in the record. Table 3 supplies the common six-group comparison.

Model	Groups scored/7	Mean RMSE	Maximum RMSE	Median R^2
Ridge	7	15.99	23.05	-0.774
Random forest	7	16.86	48.86	0.129
Histogram gradient boosting	7	17.40	56.61	0.287
Gaussian process	7	13.37	27.91	0.171
Perceptron	6	14.60	46.45	0.455

The MLP scores 0.795 on the five chalcopyrite cases held out together and -10.2 on the two gold-in-pyrite cases, with recovery RMSE of 6.72 and 46.45 points respectively. Over the common six groups, gradient boosting has mean RMSE 10.87 points and the perceptron 14.60 points. The smaller neural case-fold mean does not persist in this grouped comparison. Per-case and per-group evidence is required to locate these differences; an aggregate does not remove the rejected transfer or the poor gold-group result.

5.4 Energy and upgrade targets

The one-case-out median R^2 for total specific energy ranges from 0.652 (ridge) to 0.952 (gradient boosting); under ore-group holdout it ranges from 0.617 (ridge) to 0.907 (Gaussian process). Log-upgrade median R^2 is negative for every model under group holdout, so the recovery transfer result does not generalize to grade-upgrade prediction. The held-out protocols and target definitions remain separate in the record.

5.5 Interval coverage

Nominal 95% Gaussian-process recovery intervals cover 90.7% of interpolation rows and 82.9% of pooled case-held-out rows. The worst case-fold coverage is 35.5%, and four case folds cover fewer than 80% of their rows. These intervals use the GP's own mean and predictive standard deviation. Coverage is target- and split-specific; the authored-plant split does not establish calibrated intervals on real plants.

5.6 Reconstruction guard

On the held-out interpolation experiment, the autoencoder flags 1.47% of 612 in-envelope states and accepts 15.51% of 11,016 upward-shift probes at half a training span. Its per-feature and per-distance acceptance rates remain in `learning.json`. This validation guard differs from the final guard refitted for deployment. A low reconstruction error is a screening signal, not a process constraint, a guarantee of physical safety or a substitute for the operating contract.

6 Kinetic, optimization and sensitivity studies

The engine produces kinetic, optimization and sensitivity records for each variant. Kinetic fits, sensitivity and unscreened optimization use the mechanistic engine. Screened optimization additionally consumes the exported neural surrogate, reconstruction guard and Gaussian-process interval.

For the paired search comparison, the recorded evaluation counter counts distinct trial states in each fixed-weight multistart search's cache; a state shared across starts is counted once. Screened and unscreened runs use separate caches, and refused trial outcomes remain recorded. These counters are captured before the subsequent warm-started weight-path study. They exclude uncached base/endpoint audits, model training and screening inference, so they are a bounded search-accounting measure rather than a tally of every simulator invocation or a wall-clock benchmark.

- **Kinetic lumping.** Five lumped models fitted to a virtual batch test of the rougher feed, projected to the bank under the same residence distribution and compared with the exact bank recovery by true flotation, over 88 flotation states: mean absolute projection errors of 1.27 points for the gamma model, 1.81 for Kelsall, 2.80 for Klimpel, 4.76 for the stretched exponential and 7.56 for first order (Polat and Chander 2000; Vinnett and Waters 2025). The first-order projection falls below the exact result at every nominal state: a batch curve that is a mixture of rates is not one exponential.
- **Constrained optimization.** A generalized pattern search with a progressive barrier from six starts (Torczon 1997; Audet and Dennis 2006, 2009) finds a point within the grade, power and water constraints for 94 of the 96 variants; the two exceptions are the magnetite circuit's harder ore and higher throughput, where the grind is the only decision and the mill is already at installed power. The learned lane screens the search step (Booker et al. 1999): candidates are ranked by the perceptron's prediction and reach the engine only where the guard accepts their state and the Gaussian process's 95% half-width on recovery is within 5 points, a novelty measure for the perceptron rather than an interval on what it ranks; the poll and the optimum stay engine results. The screen did not save engine work: over the 72 screened variants it used 26,367 engine evaluations versus 22,487 without screening, 17.25% more. It saved calls in none, used more in 68 variants and tied in four. Mean absolute recovery error averaged 0.66 percentage points across the 68 variants with evaluated proposals; four variants have no proposal-error estimate rather than zero error. Across 5,811 evaluated proposals its mean was 0.71 points, median 0.48 and maximum 38.16. Both procedures had the same terminal outcome in 65 variants: 63 identical feasible decisions and two matching infeasible outcomes. Seven feasible pairs selected different decisions, with recovered-metal differences no greater than 0.002653 t/h (0.017%). Five favored the unscreened decision and two the screened decision.
- **Sensitivity.** Sobol indices of four uncertain ore properties at each nominal state (Saltelli et al. 2010; Herman and Usher 2017): floatability drives recovery in ten cases, liberation size drives concentrate

grade in seven and the head grade in the other five, the work index drives grinding energy and the head grade drives recovered metal in all twelve.

7 Measured-data comparisons

The authored circuit parameters are not fitted to these datasets. Three measured-data studies are evaluated separately; GeoMet assays also supply inputs to an explicit mechanistic comparison.

Particles. The HZDR constructed-case workbook (Pereira et al. 2021; RODARE 336, CC BY 4.0) keeps its training sheet (68 008 particles with A/B labels) and its test sheet (29 147 particles with constructed probabilities and the source authors' predictions, but no observed labels). A sparse logistic model per case and a shared network (4, 32, 32, 4) are scored against the constructed probabilities:

Case	Published reference	L1 logistic	Network
1	0.0257	0.0131	0.0138
2	0.0175	0.0157	0.0181
3	0.0207	0.0171	0.0143
4 (28 484 complete rows)	0.0371	0.1919	0.0226

The linear model beats the network in cases 1 and 2 and fails only in case 4, where the network does not.

Locked-cycle tests. 52 GeoMet locked-cycle copper recoveries from 29 drill holes of one deposit (Hoffimann et al. 2022; archived dataset 7051975, CC BY 4.0), predicted from five assays by the training mean, ridge, a random forest and a Gaussian process, under five folds of whole holes and three spatial zones. RMSE ranges from 5.09 to 5.51 points (whole holes) and 5.15 to 5.69 (zones). A paired bootstrap over complete holes (2000 resamples) shows how little separates them. On the five fixed hole folds ridge beats the training mean by 0.42 points (95% interval 0.02 to 0.82), but that partition sits at the 98.5th percentile of 200 random hole partitions; averaged over them the gain is 0.17 points (-0.39 to 0.71) and under leave one hole out 0.22 (-0.36 to 0.79), each interval widened for the six model pairs, and under spatial zones no model separates either. The source has no grind, reagent or residence information, so it cannot calibrate the engine's controls.

The engine on the same samples. The 52 samples also run through the engine as inputs, each on its own assays in the soft copper porphyry's circuit at that case's operating point; nothing in the engine is fitted to them. The work index comes from the nearest comminution sample in the hole (42 samples) or the deposit median (10), and spans 15.2 to 26.1 kWh/t against the case's 11.0. Most of the copper is bornite and chalcocite, allocated by a sulphur-limited normative mineralogy, one choice in a family of allocations; the two minerals float at declared ratios to chalcopyrite (0.8, and chalcocite at 1.5 times bornite, a choice the GeoMet data do not bound). Every sample runs the mill at installed power, and the engine's recovery falls short of the locked-cycle test by 22.6 points on average (RMSE 24.1 points; 2.9 to 41.1 points below). That gap is mostly the host circuit's size: the samples are harder than the circuit's design ore, so at 720 t/h the product is coarse. With the circuit sized for the case's 150 µm target the gap is +0.2 points, and it runs from +10.0 to -20.8 points as the assumed laboratory grind goes from 75 to 300 µm, which the open data do not give. At each sample's own target-grind throughput it is +4.0 points, of which 3.8 come from the longer flotation residence at the lower throughput rather than from the grind. At every assumed grind the engine carries no information about which sample recovers more (Pearson r between 0.03 and 0.06). It is a comparison, not a calibration: the circuit, the breakage, the liberation and the flotation are authored for another ore, and a locked-cycle test is not a plant.

Plant hours. One iron-ore plant's reverse flotation record (Kaggle dataset 6294, CC0; 737,453 rows) gives hourly laboratory silica beside 21 feed, reagent, pulp and column sensors. Dropping the 310 hours whose silica label was interpolated leaves 3,701 pairs of consecutive hours. Scored on three future windows after at least 24 hours of embargo, the previous assay alone forecasts the next hour's silica with a mean absolute error of 0.464 points, and the best sensor-only model, ridge, is 0.001 points below the training mean's 0.766. In 461 of those pairs the laboratory value is one carried over unchanged for three or more consecutive hours; without

them the previous assay's error is 0.510 points, still the lowest, and ridge is 0.014 points above the training mean. The engine has no reverse cationic flotation family, so these hours are shown and never simulated.

8 Discussion and limitations

The test bounds transfer between authored plants. A thirteenth authored plant is not a real one: the engine is a steady-state model with authored parameters, no froth stability, no dynamics and no prices, and its twelve plants share its structural assumptions. That is a feature of the design, since the teacher's answer is exact and reproducible, and a limit of the conclusion, since a real plant can depart from all twelve at once.

Four interpretation limits emerge from this simulator study. First, interpolation and transfer answer different questions: a good interpolation score coexists with a rejected transfer and a large finite error under another seed. Second, the held-out unit changes the evidence. Sharing a dominant mineral does not make whole-case and whole-group tests interchangeable; the two gold cases expose a large grouped error. Third, admissibility diagnostics change the scored denominator. A lower conditional mean cannot be presented as an improvement over a complete-design mean, and even a matched comparison remains conditional on the shared admissible folds. Fourth, reconstruction errors miss some controlled feature shifts, so their acceptance rates need the feature, distance and tested population beside them. These observations motivate a measured study on real operating periods; they do not prove the same rankings or rates on a plant.

A calibrated study would replace the authored targets by measured metallurgy across operating periods, calibrate the engine on some periods and hold out an entire period and an ore family, and score both point error and whether the surrogate preserves the ranking of operating decisions.

9 Reproducibility

The repository is https://github.com/fsantibanezleal/CAOS_OreFlow (MIT). The engine, the bake, the records, the browser port, the service and the documentation are versioned together; the source, records and computational environment are versioned with the software release. `./scripts/setup.ps1` builds the environments and `./scripts/precompute.ps1` regenerates every record. On a workstation with 32 logical cores and an RTX 4070 Laptop GPU, the pipeline took 5643 s for the cases on 8 workers and 4813 s for the learned lane. These are run-specific wall times. The final MLP fit used CUDA for a simulator study over authored cases; this computation does not establish plant-calibrated performance. Independent evaluation folds and neural seed runs can execute in bounded spawned workers. External checkpoints are accepted only under the same scientific source, JSON constants, release version, case definitions, operating contract, settings, dependency versions and compute-runtime fingerprint. Each checkpoint has a verified JSON checksum and scientific diagnostics; corrupt or foreign records fail rather than being silently reused. Execution metadata distinguishes planned, computed and resumed jobs. Final deployed-model fitting and ONNX parity checks remain part of the completed bake. Checkpoint reuse within that identical candidate is distinct from the development-only reuse of an earlier release's learning record, which is rejected for publication. `./scripts/smoke.ps1` runs the checks: the guards, the Python and frontend suites and the build, with the counts of each release in `docs/release-verification.md`. The workbench at https://oreflow.ml.fasl-work.com runs the engine in the browser on any state of any case.

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